MA'ADEN معادن
Saudi Arabian Mining Company شركة التعدين العربية السعودية

**Project** Mansourah Massarah Gold Project
**Customer:** MA'ADEN GOLD AND BASE METALS (MA'ADEN)

**Title:** CALCULATION DOCUMENT
49EE PROCESS PLANT SEWERAGE NETWORK
DESIGN OF LIFTING STATION

**Document Number:** MD-216-49EE-EG-CV-CAL-1059
**Revision:** D01
**Revision Date:** 26.12.2023
**Status:** AS BUILT

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| Outotec Document Number: | 900707 | GAU01 | AG90 | CED01 | 50005 | OU100706312 | 2 |
| L&T Document Number: | O19045 | 910 | C | K | 24363 | 001 | 2 |

**REVISION HISTORY**

| Rev | Date | Revision Text | Prepared By | Reviewed By | Approved By | Employer |
|---|---|---|---|---|---|---|
| A01 | 01.12.2020 | FIRST ISSUE | APEX | BJB | SJM | |
| C01 | 09.01.2021 | REVISED AS PER EMPLOYER COMMENT DATED 29.12.2020 | APEX | BJB | SJM | |
| D01 | 26.12.2023 | AS BUILT | APEX | BJB | SJM | |
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## DEFINITIONS AND ABBREVIATIONS

| | | |
|---|---|---|
| Document | : | Information and the medium on which it is contained (ISO 9000:2015) |
| EMPLOYER | : | 'EMPLOYER' means MA'ADEN GOLD AND BASE METALS (MGBM) and the PMC American Bechtel Inc. or their representatives |
| Contractor | : | Contractor' means consortium composed of LARSEN & TOUBRO LIMITED, Saudi Arabia AND OUTOTEC SAUDI ARABIA LLC, Saudi Arabia (for IK Contract) and LARSEN & TOUBRO LIMITED – ECC Construction Group – (SHJ BR) and OUTOTEC ENGINEERING DMCC, Dubai (for OOK Contract). |
| 'PMT' | : | Ma'aden Project Management Team |
| 'MMGP Project' | : | Mansourah Massarah Gold Processing Project |
| Attribute | : | Information element that describes a document or tag. |
| Supplier | : | An entity (supplier, consultant, sub-contractor) under the direct project management of Contractor |
| Sub-Supplier | : | A sub entity (supplier, consultant, sub-contractor) under the direct project management of Contractor. |
| QA | : | Quality Assurance |
| QC | : | Quality Control |
| Works | : | Full scope of the EPC Contract including the engineering, procurement, construction, commissioning and associated activities |


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## 1. INTENT

This document is to Prepare for design of **Lifting station** of process plant sewerage network. The analysis & design is carried out in accordance with project specifications and codes & standards. The design calculation ensures the safety and serviceability of the structure.

## 2. EXECUTIVE SUMMARY

The Saudi Arabian Mining Company (Ma'aden) has announced that its wholly-owned subsidiary, Ma'aden Gold and Base Metals Company, have issued a Notice of Award to Outotec and Larsen & Toubro Consortium to complete the engineering, procurement, construction, pre-commissioning, commissioning and start-up assistance of the Mansourah & Massarah (MMGP) Gold Project.

MMGP project will become Ma'aden's largest ever gold project, producing an average of 250,000 ounces of gold per year over the life of mine. Mansourah-Massarah is situated within the Central Arabian Gold Region (CAGR), approximately 320km northeast of Taif towards Riyadh and about 35 km southeast of Hufairah, which is a small village situated on Jeddah - Riyadh highway. Mansourah - Massarah gold deposits are located 22°38'6.40"N; 42°44'22.90"E, and 22°42'1.40"N; 42°41'58.40"E respectively, approximately 537 km east of Jeddah and 77 km southeast of the town of Zalim.

The plant throughput capacity is 4000000t/a and is required to operate on a continuous basis at nominally 24 hours per day and 7 days a week.

## 3. SCOPE

The analysis & design of wall and base raft have been carried out in Excel sheet. For wall and base raft design, force values are calculated in excel sheet.


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# 4. DESIGN REQUIREMENTS

## 4.1 Design

Design of concrete structures and foundations shall be in accordance with SBC-304, ACI 318M and API 650.

## 4.2 Materials

Concrete Strength of 28 days

- Foundation and Structural Members: 35 MPa
- Blinding Concrete below foundation bases: 15 MPa

Reinforcement Grade

- Deformed Bar: ASTM A 615 / A 615M Grade 60 (fy = 420MPa)
- Welded Wire Fabric: ASTM A 185(Plain) / A 497(Deformed)

## 4.3 Minimum Concrete Cover

| Concrete Exposure | Member | Minimum Cover(mm) |
|---|---|---|
| Contact with ground | Wall & Base raft | 50 |
| Exposed to weather | Top slab and beam | 25 |

## 4.4 Minimum Reinforcing Steel (Rebar) Size

The minimum rebar size for use as tension, compression reinforcement shall be 12 mm, except where welded wire fabric is used. For ties and spirals the minimum rebar size shall be 10 mm.

## 4.5 Allowable Soil Bearing Capacity

The net safe bearing capacity of soil for foundations design has been considered as 1144 kN/m$^2$ at 1.5m depth below EGL refer Table 11 - BH-22 of Geotechnical Report (MD-216-3000-EG-CV-RPT-1008)


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## 5. DESIGN CALCULATION FOR LIFTING STATION


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## Wall Section at Bottom

**Materials**

Compressive Strength of Concrete at 28th days (fc') = 35 Mpa= 35 N/sqmm
Yield Strength of Steel (fy) = 420 Mpa= 420 N/sqmm

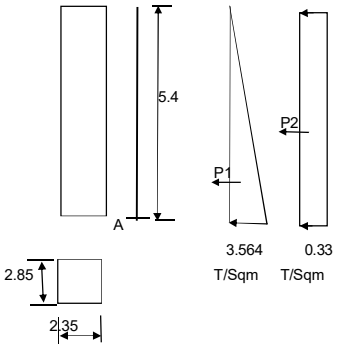
**Unit Weight**

Reinforced Cement Concrete(RCC)= 2.4 T/Cum
Plain Cement Concrete(PCC)= 2.3 T/Cum
Soil = 2.0 T/Cum

Co efficient of Active Earth Pressure(ka) = 0.33
Surcharge considered = 1.0 T/Sqm

**Vertical Wall**

Section at Bottom(Considering Horizontal Bending)




| | | | |
|---|---|---|---|
| Horizontal SF due to Soil at A, P1 = | 5.08 T | (wl/2) | |
| Horizontal SF due to Surcharge at A, P2 = | 0.47 T | (wl/2) | |
| Horizontal Supt Moment at A due to Soil, M1 = | 2.412 TM | | ($wl^2$/12) |
| Horizontal Supt Moment at A due to Surcharge, M2 = | 0.223 TM | | ($wl^2$/12) |
| Design Moment at A, Mu= | 3.275 TM | | (1.2M1+1.7M2) |
| Design Shear at A, SFu= | 6.89 T | | (1.2P1+1.7P2) |
| Wall Thickness(D) = | 400.00 mm | | |
| Clear Cover to Reinf = | 50.00 mm | | |
| Dia of Reinforcement = | 16.00 mm | (Both Horizontal & Vertical) | |
| Effective Depth(d) = | 342.00 mm | | |
| Width(b) = | 1000.00 mm | | |

**Calculation of flexural reinforcement**

$M_{lim} = Ø0.85f'_c\beta_1(c/d)[1-\beta_1/2(c/d)]bd^2$
= 798.586 KNm > Mu

Ø = 0.9
c/d = 0.003/(0.003+0.005) = 0.375
$\beta_1$ = 0.8

$A_{st} = (0.85f'_cbd/f_y)[1\sqrt{1-(2.35M_u/Ø f'_cbd^2)}\ ]$
= 254.32 sqmm <$A_{stmin}$

$(0.85f'_cbd/f_y)$ = 24225.000
$\sqrt{1-(2.35M_u/Ø f'_cbd^2)}$ = 0.98950

Dia of Reinf provided = 16.00 mm mm
Spacingof Reinf provided = 150 mm c/c
$A_{st}$ provided = 1340.41 sqmm ok

$A_{st,min}$ (greater than)= a)0.0018*420*$A_g$/$f_y$ or b)0.0014$A_g$
= 720 or 560 sqmm

**Shear Check**

Reinforcement ratio = $\rho_w$ = 0.00392
Effective Depth = d = 342.00 mm
Moment = Mu = 32.7 KNm
Shear Force = $V_u$ = 68.939 KN
λ = 1 for normal concrete
Φ = 0.75 for shear
Allowable shear strength = ΦVc = 255.10 KN, Hence OK

Lesser of the following :

1) $V_c = (0.16\lambda\sqrt{f'_c} + 17\rho_w(V_ud/Mu))b_wd$
= 340134.54 N
ΦVc = 255.10 KN

2) $V_c = (0.16\lambda\sqrt{f'_c} + 17\rho_w)b_wd$
= 1144213.31 N
ΦVc = 858.16 KN

3) $V_c = (0.29\lambda\sqrt{f'_c})\ b_wd$
= 586756.7929 N
ΦVc = 440.07 KN


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**Crack Width Check :**

| | | | | | |
|---|---|---|---|---|---|
| $E_s$ = | 200000 | Mpa | λ = 1 | | |
| $E_c$ = | 27805.57 | Mpa | n = 7.1928 | | |
| $f_r$ = | 3.67 | Mpa | *[ Modulus of rupture of concrete ( ACI 318, eq 9-10)]* | | |
| $I_g$ = | 0.0053 | $m^4$ | $y_t$ = | 0.2 | m |
| $M_{cr}$ = | 97.813 | kN-m | | | |
| $M_{srv}$ = | 32.746 | kN-m | **O.K.** | | |
| B = | 0.1037 | $mm^{-1}$ | | | |
| Neutral axis depth, c = | 72.137 | mm | | | |
| MOI of Cracked section, $I_{cr}$ = | 8.27E+08 | $mm^4$ | | | |
| Steel stress, $f_s$ = | 76.834 | Mpa | | | |
| | 76.834 | Mpa | | | |
| Concrete stress, $f_c$ = | 2.855 | Mpa | | | |
| $s_{max}$ = | 1259.806 | mm | **O.K.** *[ Max Spacing ( ACI 318, eq 10-4)]* | | |
| b = | 1.2149 | | $d_c$ = | 74 | mm |
| A = | 22200 | $mm^2$ | | | |
| w = | 0.098 | mm | *[ ACI 224.1R-07, eq 1-1 ]* | | |
| $w_{limit}$ = | 0.1 | mm | **O.K.** | | |


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**Wall Section at Top**

**Materials**

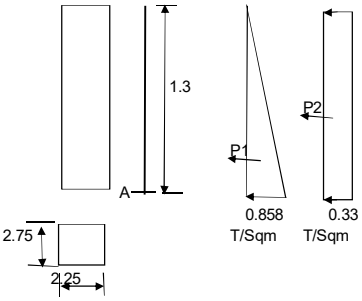
| | | |
|---|---|---|
| Compressive Strength of Concrete at 28th days (fc') = | 35 Mpa= | 35 N/sqmm |
| Yield Strength of Steel (fy) = | 420 Mpa= | 420 N/sqmm |

**Unit Weight**

| | |
|---|---|
| Reinforced Cement Concrete(RCC)= | 2.4 T/Cum |
| Plain Cement Concrete(PCC)= | 2.3 T/Cum |
| Soil = | 2.0 T/Cum |
| Co efficient of Active Earth Pressure(ka) = | 0.33 |
| Surcharge considered = | 1.0 T/Sqm |

**Vertical Wall**

Section at Top(Considering Horizontal Bending)




| | | | |
|---|---|---|---|
| Horizontal SF due to Soil at A, P1 = | 1.18 T | (wl/2) | |
| Horizontal SF due to Surcharge at A, P2 = | 0.45 T | (wl/2) | |
| Horizontal Supt Moment at A due to Soil, M1 = | 0.541 TM | | ($wl^2/12$) |
| Horizontal Supt Moment at A due to Surcharge, M2 = | 0.208 TM | | ($wl^2/12$) |
| Design Moment at A, Mu= | 1.002 TM | | (1.2M1+1.7M2) |
| Design Shear at A, SFu= | 2.19 T | | (1.2P1+1.7P2) |
| Wall Thickness(D) = | 275.00 mm | | |
| Clear Cover to Reinf = | 50.00 mm | | |
| Dia of Reinforcement = | 12.00 mm | (Both Horizontal & Vertical) | |
| Effective Depth(d) = | 219.00 mm | | |
| Width(b) = | 1000.00 mm | | |

**Calculation of flexural reinforcement**

$M_{lim} = Ø0.85f'c\beta_1(c/d)[1-\beta_1/2(c/d)]bd^2$ = 327.460 KNm > Mu

Ø = 0.9

c/d = 0.003/(0.003+0.005) = 0.375

$\beta_1$ = 0.8

$A_{st} = (0.85f'_cbd/f_y)[1-\sqrt{1-(2.35M_u/Ø f'_cbd^2)}]$ = 121.41 sqmm <$A_{stmin}$

$(0.85f'_cbd/f_y)$ = 15512.500

$\sqrt{1-(2.35M_u/Ø f'_cbd^2)}$ = 0.99217

Dia of Reinf provided = 12.00 mm mm

Spacingof Reinf provided = 150 mm c/c

$A_{st}$ provided = 753.98 sqmm ok

$A_{st,min}$ (greater than)= a)0.0018*420*$A_g/f_y$ or b)0.0014$A_g$ = 495 or 385 sqmm

**Shear Check**

| | | | |
|---|---|---|---|
| Reinforcement ratio = $\rho_w$ = | 0.00344 | | |
| Effective Depth = d = | 219.00 mm | | |
| Moment = Mu = | 10 KNm | | |
| Shear Force = $V_u$ = | 21.871 KN | | |
| λ = | 1 | for normal concrete | |
| Φ = | 0.75 | for shear | |
| Allowable shear strength = ΦVc = | 160.07 KN, Hence | OK | |

Lesser of the following :

1) $V_c = (0.16\lambda\sqrt{f'_c} + 17\rho_w(V_ud/Mu))b_wd$ = 213423.96 N

ΦVc = 160.07 KN

2) $V_c = (0.16\lambda\sqrt{f'_c} + 17\rho_w)b_wd$ = 730924.01 N

ΦVc = 548.19 KN

3) $V_c = (0.29\lambda\sqrt{f'_c})\,b_wd$ = 375730.227 N

ΦVc = 281.80 KN


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**Crack Width Check :**

| | | | | | |
|---|---|---|---|---|---|
| $E_s$ = | 200000 | Mpa | λ = 1 | | |
| $E_c$ = | 27805.57 | Mpa | n = 7.1928 | | |
| $f_r$ = | 3.67 | Mpa | *[ Modulus of rupture of concrete ( ACI 318, eq 9-10)]* | | |
| $I_g$ = | 0.0017 | $m^4$ | $y_t$ = | 0.1375 | m |
| $M_{cr}$ = | 46.232 | kN-m | | | |
| $M_{srv}$ = | 10.024 | kN-m | **O.K.** | | |
| B = | 0.1844 | $mm^{-1}$ | | | |
| Neutral axis depth, c = | 43.615 | mm | | | |
| MOI of Cracked section, $I_{cr}$ = | 1.94E+08 | $mm^4$ | | | |
| Steel stress, $f_s$ = | 65.024 | Mpa | | | |
| | 65.024 | Mpa | | | |
| Concrete stress, $f_c$ = | 2.248 | Mpa | | | |
| $s_{max}$ = | 1511.324 | mm | **O.K.** | *[ Max Spacing ( ACI 318, eq 10-4)]* | |
| b = | 1.3193 | | $d_c$ = | 68 | mm |
| A = | 20400 | $mm^2$ | | | |
| w = | 0.087 | mm | *[ ACI 224.1R-07, eq 1-1 ]* | | |
| $w_{limit}$ = | 0.1 | mm | **O.K.** | | |


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**Base Raft**

**Materials**

| | | |
|---|---|---|
| Compressive Strength of Concrete at 28th days (fc') = | 35 Mpa= | 35 N/sqmm |
| Yield Strength of Steel (fy) = | 420 Mpa= | 420 N/sqmm |

**Unit Weight**

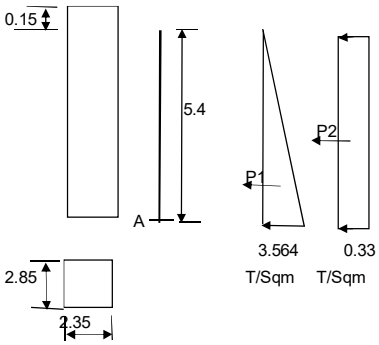
| | |
|---|---|
| Reinforced Cement Concrete(RCC)= | 2.4 T/Cum |
| Plain Cement Concrete(PCC)= | 2.3 T/Cum |
| Soil = | 2.0 T/Cum |
| Co efficient of Active Earth Pressure(ka) = | 0.33 |
| Surcharge considered = | 1.0 T/Sqm |

**Base Raft**

| | |
|---|---|
| Inside $L_1$ = | 2.5 M |
| Inside $B_1$ = | 2.0 M |
| Base raft Proj= | 0.15 M |
| Wall thickness at Bot= | 0.35 M |
| Wall thickness at Top= | 0.25 M |
| Total L= | 3.5 M |
| Total B= | 3.0 M |
| Thickness of Raft= | 0.35 M |
| Base raft c/c of wall Ly= | 2.85 M |
| Base raft c/c of wall Lx= | 2.35 M |
| 250 thk Wall height above Gr = | 0.15 M |
| 251 thk Wall height below Gr = | 0.9 M |
| 350 thk Wall height below Gr = | 3.4 M |
| Total Height of Wall= | 4.45 M |

**Total Weight of Pit**

| | | |
|---|---|---|
| Weight of 350 thk Wall= | 29.70 T | |
| Weight of 250 thk Wall= | 4.38 T | |
| Weight of Raft= | 8.82 T | |
| Weight of Soil= | 16.00 | |
| Top Cover(Chqd Plate) = | 1.05 T | (@100 kg/sqm) |
| Total = | 59.95 T | |
| Pressure at Bottom = | 5.71 T/sqm | ok |
| Span ratio K=Ly/Lx = | 1.21 | |
| Short Span BM Mx = $\infty_x wL_x^2/8$ = | 1.76 TM | $\infty_x$=0.446 from Reynolds Handbook |
| Long Span BM My = $\infty_y wL_y^2/8$ = | 1.06 TM | $\infty_y$=0.183 from Reynolds Handbook |
| Shear Force SF = | 8.21 T | |




| | | |
|---|---|---|
| Design Moment at Supt, Mu= | 2.109 TM | (1.2*BM) |
| Design Shear at Supt, SFu= | 9.86 T | (1.2*SF) |
| Raft Thickness(D) = | 400.00 mm | |
| Clear Cover to Reinf = | 50.00 mm | |
| Dia of Reinforcement = | 16.00 mm | |
| Effective Depth(d) = | 342.00 mm | |
| Width(b) = | 1000.00 mm | |


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**Calculation of flexural reinforcement**

$M_{lim} = Ø0.85f'_c\beta_1(c/d)[1-\beta_1/2(c/d)]bd^2$ = 798.586 KNm > Mu

Ø = 0.9

c/d = 0.003/(0.003+0.005) = 0.375

$\beta_1$ = 0.8

$A_{st} = (0.85f'_cbd/f_y)[1-\sqrt{1-(2.35M_u/Ø f'_cbd^2)}]$ = 163.51 sqmm < $A_{stmin}$

$(0.85f'_cbd/f_y)$ = 24225.000

$\sqrt{(2.35M_u/Ø f'_cbd^2)}$ = 0.99325

Dia of Reinf provided = 16.00 mm mm

Spacingof Reinf provided = 150 mm c/c

$A_{st}$ provided = 1340.41 sqmm ok

$A_{st,min}$ (greater than)= a)0.0018*420*$A_g/f_y$ or b)0.0014$A_g$

= 720 or 560 sqmm

**Shear Check**

Reinforcement ratio = $\rho_w$ = 0.00392

Effective Depth = d = 342.00 mm

Moment = Mu = 21.1 KNm

Shear Force = $V_u$ = 98.564 KN

λ = 1 for normal concrete

Φ = 0.75 for shear

Allowable shear strength = ΦVc = 270.11 KN, Hence OK

Lesser of the following :

1) $V_c = (0.16\lambda\sqrt{f'_c} + 17\rho_w(V_ud/Mu))b_wd$

= 360144.86 N

ΦVc = 270.11 KN

2) $V_c = (0.16\lambda\sqrt{f'_c} + 17\rho_w)b_wd$

= 1144213.31 N

ΦVc = 858.16 KN

3) $V_c = (0.29\lambda\sqrt{f'_c})\,b_wd$

= 586756.7929 N

ΦVc = 440.07 KN

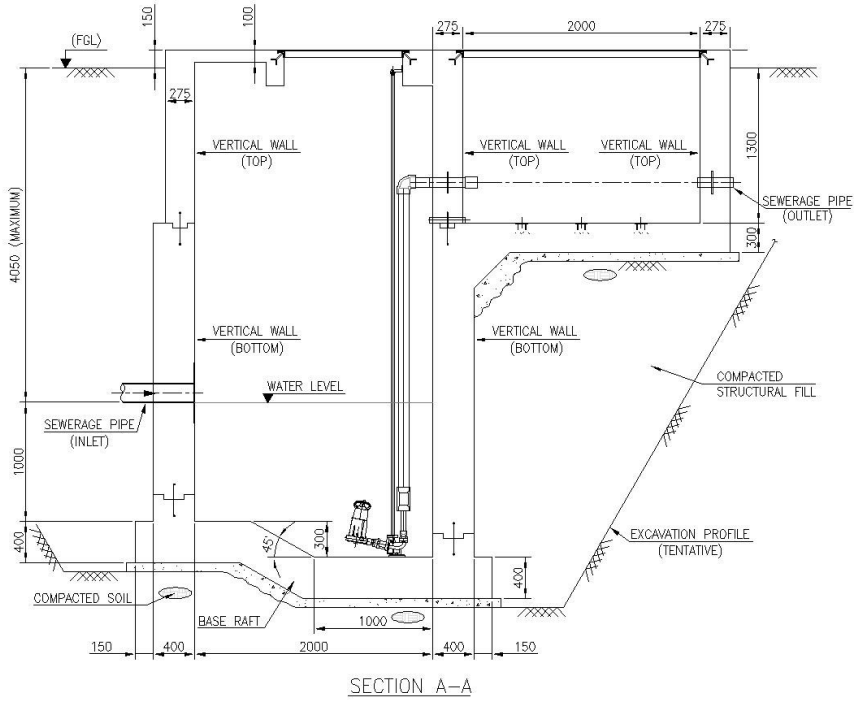

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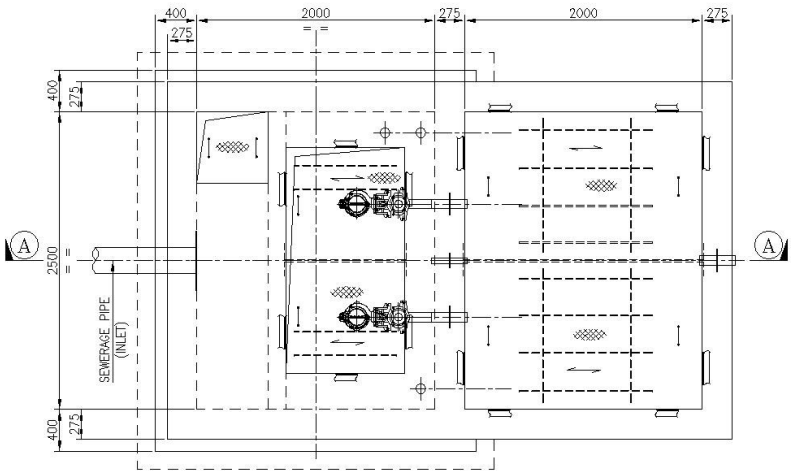
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C01

## SKETCH OF LIFTING STATION




SECTION A–A




TOP PLAN & DETAILS OF LIFTING STATION

NOTE:– ALL DIMENSIONS ARE IN MILLIMETRE.


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## REFERENCES

Reference is made to codes, standards and specifications as indicated in Civil, Structural and Architectural Design Criteria (Doc. No. MD-216-1000-EG-ST-CRT-1001) Clause 4.1


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C01

# ATTACHMENT 1 COMMENT RESOLUTION LOG SHEET


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MA'ADEN معادن
Saudi Arabian Mining Company شركة التعدين العربية السعودية

BECHTEL

# Mansourah Massarah Gold Project
## Comment Resolution Log Sheet / Compliance Report

Outotec
LARSEN & TOUBRO

**PROJECT:** MANSOURAH MASSARAH GOLD PROJECT **PROJECT NO.: 216**
**STAGE:** 4 - PROJECT EXECUTION. **REVIEW LOG DOC. NO. & REV.:**
**ACONEX REF. NO. & DATE:** BECHTEL-WTRAN-007394, 29 December 2020
STATUS CODE: A - APPROVED, C - APPROVED WITH COMMENTS, R - REVISE & RESUBMIT & O - OTHERS (INFORMATION, REFERENCE, REVIEWED)

| DOCUMENT NUMBER | Area / Title | Comment No. | PMC/PMT COMMENT<br>DATE: 29 December 2020 | Comment by | EPC Response<br>DATE: 09 January 2021 | Status | Remarks |
|---|---|---|---|---|---|---|---|
| | | | **General Comments (cc)** | | | | |
| MD-216-49EE-EG-CV-CAL-1059-A01 | 49EE | 1 | Please provide some sketches for better understanding the location of each element you are designing. | Mr. Doron Golan Bechtel | Comment has been incorporated in the calculation document. | Closed | |